

COMPUTATIONAL ANALYSIS OF FLOW IN PINTLE INJECTOR

A PROJECT REPORT

Submitted to

SRM INSTITUTE OF SCIENCE & TECHNOLOGY

By

ANUSHKA MULIK (RA2211019010006)

TANUSH LAD (RA2211019010045)

ADESH LAWANDE (RA2211019010051)

in partial fulfilment for the award of the degree

Of

BACHELOR OF TECHNOLOGY

IN

AEROSPACE ENGINEERING



SRM

INSTITUTE OF SCIENCE & TECHNOLOGY
(Deemed to be University u/s 3 of UGC Act, 1956)

DEPARTMENT OF AEROSPACE ENGINEERING

SRM INSTITUTE OF SCIENCE & TECHNOLOGY

KATTANKULATHUR 603203,

CHENNAI, INDIA.

MAY 2025

SRM INSTITUTE OF SCIENCE & TECHNOLOGY

(Under Section 3 of UGC Act, 1956)

BONAFIDE CERTIFICATE

Certified that this project report titled “**COMPUTATIONAL ANALYSIS OF FLOW IN PINTLE INJECTOR**” is the bonafide work of “**ANUSHKA MULIK, TANUSH LAD & ADESH LAWANDE**”, who carried out the project work under my supervision. Certified further, that to the best of my knowledge the work reported herein does not form any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

DR. SARAVANAN G

PROJECT GUIDE

Assistant Professor

Department of Aerospace Engineering

DR. L. R. GANAPATHY SUBRAMANIAN

PROFESSOR AND HEAD,

Department of Aerospace Engineering

(Internal Examiner - 1)

(Internal Examiner - 2)

ABSTRACT

This study investigates the computational fluid dynamics (CFD) analysis of flow characteristics in pintle injectors for liquid rocket engines. Pintle injectors, which enable effective throttling capabilities through a movable pintle mechanism, are critical components that influence combustion stability and performance. Multiple 3D models with varying pintle gap distances are designed and simulated to evaluate the effect of geometrical parameters on flow behaviour. The computational approach utilized the Lagrangian method to model liquid-gas interactions, with realizable k-epsilon turbulence modeling employed to capture complex flow patterns. Results demonstrated that a 3 mm pintle gap provided optimal performance parameters in terms of pressure distribution, velocity profiles, and spray characteristics. Additionally, a 40° pintle tip angle were identified as the optimal configuration for achieving desired atomization performance. While attempts to validate these computational results through 3D-printed prototypes are hindered by manufacturing limitations that led to injector choking, the simulation data provided valuable insights into the mixing phenomena and spray formation processes. This research extends our understanding of how injection conditions and geometric parameters influence the performance of pintle injectors, with important implications for throttleable liquid rocket engine design.

ACKNOWLEDGEMENT

We are deeply grateful to our guide, **Dr. Saravanan G (Assistant Professor)**, for his unwavering support, insightful suggestions, and critical feedback at every stage of this project. His encouragement motivated us to tackle the challenges of CAD design, 3D printing trials, and CFD simulation with confidence and rigor.

We also extend our sincere thanks to the three project team members — **Anushka Mulik, Tanush Lad, and Adesh Lawande** —for their collaborative spirit and dedication. Together, we handled all aspects of the work independently: designing and iterating the pintle injector geometry, troubleshooting the 3D-printed prototypes, and conducting detailed CFD analyses in ANSYS Fluent. Each member contributed unique skills—ranging from SolidWorks modeling to mesh generation and post-processing—that were essential to achieving our results.

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	III
	ACKNOWLEDGEMENT	IV
	TABLE OF CONTENTS	V
	LIST OF TABLES	VII
	LIST OF FIGURES	VIII
	LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE	IX
1	INTRODUCTION	1
2	LITERATURE SURVEY	4
3	OBJECTIVES AND METHODOLOGY	13
4	NUMERICAL ANALYSIS / SIMULATION	19
	4.1 GEOMETRY AND CAD DESIGN	19
	4.2 MESH SETUP	22
	4.3 BOUNDARY CONDITIONS	24
	4.4 SOLVER CONFIGURATION	25
	4.5 SIMULATION CASES	26
5	RESULTS AND DISCUSSION	30
	5.1 VELOCITY FIELD ANALYSIS	30
	5.1.1 PEAK AXIAL VELOCITY	30
	5.1.2 JET CORE WIDTH AND DECAY	30
	5.1.3 RADIAL VELOCITY COMPONENTS	32
	5.2 PRESSURE DISTRIBUTION AND LOSSES	32
	5.2.1 PRESSURE DROP ACROSS THE GAP	32
	5.2.2 CHAMBER RECOVERY	32
	5.2.3 LOCALIZED LOW-PRESSURE ZONES	33
	5.3 SPRAY SPREAD AND MIXING POTENTIAL	33
	5.3.1 SPRAY HALF-ANGLE	33
	MEASUREMENT	

	5.3.2 PENETRATION VS DISPERSION	33
	TRADE-OFF	
	5.3.3 MIXING-QUALITY PARAMETER (NK)	34
	5.4 IDENTIFICATION OF OPTIMAL CONFIGURATION	34
	5.5 COMPARISON WITH LITERATURE	34
	5.6 PRACTICAL IMPLICATIONS AND LIMITATIONS	35
6	CONCLUSIONS	37
7	FUTURE ENHANCEMENTS	43
	REFERENCES	49
	APPENDIX	53

LIST OF TABLES

TABLE NO.	TITLE	PAGE NO.
4.1	INFLUENCE OF PINTLE GAP SIZE ON SPRAY PATTERN UNIFORMITY.	27
4.2	EFFECT OF PINTLE GAP ON VELOCITY, PRESSURE DROP, AND SPRAY ANGLE	30

LIST OF FIGURES

FIGURE NO.	TITLE	PAGE NO.
1.1	SCHEMATIC DIAGRAM OF PINTLE INJECTOR	1
1.2	CROSS-SECTION DIAGRAM OF PINTLE INJECTOR	2
4.1	DIMENSIONAL DIAGRAM OF THE PINTLE INJECTOR	21
4.2	SECTION VIEW OF OUTER PART OF PINTLE INJECTOR	21
4.3	CAD OF OUTER PART OF PINTLE INJECTOR	21
4.4	SECTION VIEW OF INNER PART OF PINTLE INJECTOR	21
4.5	ISOMETRIC VIEW OF INNER PART OF PINTLE INJECTOR	21
4.6	SECTION VIEW OF OUTER PART OF PINTLE INJECTOR (ITERATION-2)	22
4.7	ISOMETRIC VIEW OF OUTER PART OF PINTLE INJECTOR (ITERATION-2)	22
4.8	ASSEMBLED VIEW OF PINTLE INJECTOR (ITERATION-2)	22
4.9	VELOCITY CONTOUR OF FLOW IN PINTLE INJECTOR	27
4.10	PRESSURE CONTOUR OF FLOW IN PINTLE INJECTOR	27
5.1	PRESSURE CONTOUR OF FLOW IN PINTLE INJECTOR (ITERATION-2)	32
5.2	VELOCITY CONTOUR OF FLOW IN PINTLE INJECTOR (ITERATION-2)	32

LIST OF SYMBOLS, ABBREVIATION, AND NOMENCLATURE

SYMBOL /ABBREVIATION	DESCRIPTION
CFD	Computational Fluid Dynamics
CEA	Chemical Equilibrium Analysis
CH ₄	Methane (Fuel)
LOX	Liquid Oxygen (Oxidizer)
C _p	Specific Heat at Constant Pressure (J/kg·K)
γ (gamma)	Ratio of Specific Heats (C _p /C _v)
ρ (rho)	Density (kg/m ³)
P	Pressure (Pa or bar)
T	Temperature (K)
I _{sp}	Specific Impulse (s or m/s)
C*	Characteristic Velocity (m/s)
O/F	Oxidizer-to-Fuel Ratio
MPa	Megapascal (10 ⁶ Pascal)
mm	Millimeters
LES	Large Eddy Simulation
RANS	Reynolds-Averaged Navier–Stokes
SIMPLE	Semi-Implicit Method for Pressure-Linked Equations

INTRODUCTION

The pintle injector was pioneered by TRW Inc. in the 1950s and gained prominence through its successful implementation in the Apollo Lunar Module Descent Engine (LMDE), which safely delivered astronauts to the lunar surface. Unlike traditional injector designs that utilize fixed geometry, the pintle injector incorporates a movable central element—the pintle rod—which can be axially adjusted to vary the effective injection area. This distinctive feature allows for substantial throttling capabilities, typically ranging from 10:1 to 20:1, far exceeding the capabilities of conventional fixed-geometry injectors.



In pintle injectors, liquid propellant is typically injected radially outward through an annular orifice formed between the pintle tip and the surrounding injector body, creating a thin liquid sheet. Simultaneously, the gaseous propellant (or in some configurations, the second liquid propellant) flows axially through an outer annular passage and impinges upon this liquid sheet. This impingement process generates well-atomized spray patterns critical for efficient combustion. The resulting spray characteristics, including droplet size distribution, spray angle, and mixing efficiency, significantly influence the combustion stability, performance, and thermal management of the rocket engine.

Despite its proven reliability in various space missions, the complex fluid dynamic phenomena occurring within pintle injectors remain challenging to predict analytically. The interaction between the radially injected liquid sheet and the axially flowing gas stream involves intricate breakup mechanisms, turbulent mixing processes, and potential recirculation zones that defy simplified modeling approaches. The spatial distribution of propellants, degree of atomization, and resultant mixing efficiency are highly sensitive to geometric parameters—particularly the pintle gap distance and tip angle—as well as operational factors such as injection velocities and momentum ratios between the propellant streams.

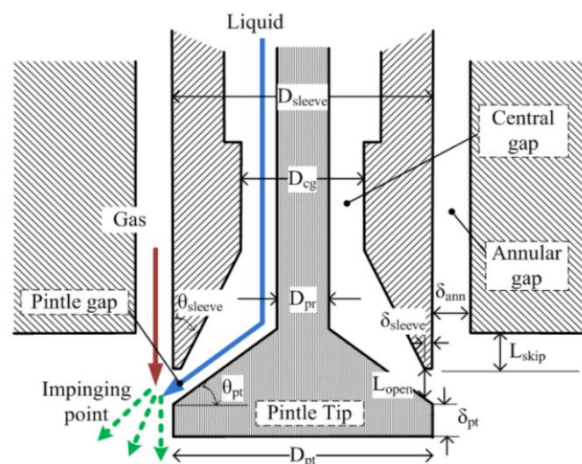


Figure 1.2 Cross-section diagram of Pintle Injector

Computational Fluid Dynamics (CFD) has emerged as an indispensable tool for understanding and optimizing the complex flow phenomena within rocket injectors. Modern CFD approaches enable detailed investigation of multiphase flows with

considerations for turbulence, phase interactions, and breakup mechanisms that would be difficult to observe experimentally. The Lagrangian approach, which tracks discrete particles or droplets within a continuous phase, has proven particularly valuable for modeling spray formation and propellant mixing in pintle injectors.

Experimental validation remains essential for verifying computational results, though it presents significant challenges due to the harsh conditions typical of rocket combustion environments. While cold-flow testing (using non-reactive simulants) provides valuable data on spray characteristics without the complications of combustion, translating these findings to actual hot-fire conditions requires careful consideration of additional factors such as heat release, gas expansion, and chemical kinetics.

The manufacturing of experimental pintle injector prototypes introduces additional complexities, particularly when utilizing advanced fabrication techniques such as additive manufacturing (3D printing). The intricate internal geometry of pintle injectors, with narrow passages and precise dimensional requirements, pushes the limits of current manufacturing capabilities. As experienced in this research, such limitations can lead to manufacturing defects that impede experimental validation efforts.

This study focuses on the computational fluid analysis of flow in pintle injectors, with particular emphasis on understanding how geometric parameters—specifically the pintle gap distance and tip angle—influence the resulting flow behavior, spray characteristics, and mixing performance. Through systematic variation of these parameters and analysis of resulting flow patterns, this research aims to identify optimal configurations for liquid rocket engine applications and provide insights into the underlying physical mechanisms governing pintle injector performance.

The significance of this research extends beyond academic interest, as optimized pintle injector designs can enable more efficient, reliable, and versatile rocket propulsion systems. The deep throttling capability of pintle injectors is particularly valuable for missions requiring precise landing maneuvers, orbital adjustments, or long-duration burns with varying thrust requirements.

CHAPTER 2

LITERATURE SURVEY

A. CFD Modeling of Pintle Injector Spray and Atomization

Son & Radhakrishnan (2013)

Son and Radhakrishnan performed a comprehensive CFD analysis on a fuel-centered pintle injector to investigate spray formation and atomization. Using a Lagrangian droplet model coupled with the Realizable $k-\epsilon$ turbulence model, they explored the effects of varying pintle opening sizes and fuel-to-gas mass flow ratios on spray characteristics. Key metrics analyzed included spray half-angle, breakup length, and Sauter Mean Diameter (SMD). They also proposed a new mixing-quality parameter, NK, to quantify atomization efficiency. Their findings showed that minor changes in pintle geometry could significantly alter spray behavior. In our study, we adopted their methodology by keeping the injection angle constant and varying the pintle gap between 2 mm and 5 mm to assess dispersion and atomization efficiency.

Kun Cai et al. (2023)

This simulation study employed the Volume of Fluid (VOF) and Large Eddy Simulation (LES) methods to assess how slit and slot configurations in pintle injectors influence atomization. It revealed that increased flow rate and larger pintle diameter improve atomization quality, while slot geometry critically affects spray distribution. The authors highlighted the importance of optimizing slot number, shape, and thickness to enhance spray uniformity. We referred to this work when analyzing how geometric adjustments to the pintle gap could affect spray patterns in our design.

Yuan Xie et al. (2022)

This paper used a multiphase CFD model combining Eulerian and Lagrangian approaches to simulate internal flow and spray development in pintle injectors. The authors investigated how nozzle geometry and inlet pressure affected droplet breakup, penetration, and spray shape. The simulation framework provided valuable insights into

internal and near-nozzle flow structures, which we replicated in our own CFD model to ensure realistic prediction of atomization behavior.

Wansheng Nie et al. (2022)

Using LES and Lagrangian particle tracking, this study analyzed how different pintle injector geometries affected spray characteristics such as droplet size distribution, spray penetration, and breakup length. The authors demonstrated that injector geometry directly influences atomization efficiency. We adopted a similar simulation methodology, focusing on how pintle gap variation influenced the overall spray structure in cold-flow conditions.

Arpit Dubey et al. (2023)

This work investigated pintle injector configurations in hybrid propulsion systems using CFD. It focused on evaluating how changes in pintle geometry affected fuel mixing and combustion performance. Although our study was limited to cold-flow conditions, the study emphasized the importance of injector design precision for optimal mixing and performance, particularly relevant for future applications involving propellant mixing in hybrid systems.

Rajiv kumar et al. (2023)

This study used both experiments and CFD simulations to evaluate how fuel injection pressure impacts spray quality, droplet size, and combustion efficiency. Higher pressures resulted in finer atomization and more stable combustion. In our simulations, we applied this understanding by varying pressure to assess its impact on spray dispersion and droplet dynamics.

B. CFD Methodology and Simulation Practices

Tsohas & Heister (2006)

Tsohas and Heister emphasized the critical role of mesh resolution in CFD simulations of coaxial injectors. They demonstrated that resolving shear layers and recirculation zones requires fine grid refinement and proper boundary-layer modeling. Their mesh-

independence study, involving inflation layers and quality checks, ensured accurate flow predictions. We followed a similar approach by conducting mesh-independence analysis in our simulations, guaranteeing that results were not sensitive to grid size.

Landrum (2003)

The paper by Landrum, Beardt, and Dingust (1995) titled "Performance Characterization of Plain-Orifice Injectors with Very Small Hole" was presented at the 31st Joint Propulsion Conference and Exhibit in San Diego. This study investigates the performance characteristics of injectors featuring very small orifices, which are pertinent to the design considerations of pintle injectors. While the full details of the paper are not readily available in the provided sources, such studies typically focus on aspects like flow dynamics, discharge coefficients, and atomization behavior of injectors with micro-scale orifices. Understanding these characteristics is crucial for optimizing injector designs to achieve desired performance metrics in propulsion systems.

Farmer & Cheng (2000)

Farmer and Cheng developed a CFD injector design framework integrating turbulence and spray breakup models. They emphasized the use of second-order upwind schemes for accurate spatial discretization and the SIMPLE algorithm for pressure-velocity coupling. We adopted similar numerical strategies in our simulations to resolve injector flow features accurately and ensure solution stability in cold-flow conditions.

C. Experimental and Empirical Studies on Injectors

Yildiz (2005)

In his 2005 study, Tolga Yıldız conducted an experimental investigation on the atomization and spray characteristics of like-doublet impinging-jet injectors. The research aimed to establish correlations between key parameters such as jet velocity, impingement angle, and spray half-angle. Utilizing cold-flow experiments with water simulants, Yıldız employed tools like a mechanical patternator, Particle Dynamics

Analyzer (PDA), and high-speed shadowgraphy to obtain both quantitative and qualitative data on aspects like flux distributions, droplet statistics, and spray morphologies. The study found that with an increase in Weber number, the spray angle gradually increases, stabilizing at a maximum of 85° when jet velocity is sufficient. These empirical relationships provided valuable benchmarks for injector performance. In our CFD design, we applied Yıldız's correlations to select a 40° pintle tip angle, ensuring a balanced spray pattern that optimizes both penetration and radial dispersion.

Talley (1998)

In his 1998 study, Talley classified various atomization regimes and proposed empirical correlations for breakup length in different injector types. These correlations serve as benchmarks for predicting spray performance, aiding in the design and optimization of injectors. In our CFD study, we utilized Talley's data to compare predicted breakup lengths for different pintle gaps, validating the physical accuracy of our simulations. This approach ensures that the simulated spray characteristics align with established empirical data, enhancing the reliability of our design predictions.

Sanjay Kumar Gupta et al. (2024)

In their 2024 study, Sanjay Kumar Gupta and colleagues conducted an experimental investigation into the spray formation of a novel pintle injector featuring a bi-swirl arrangement, developed using additive manufacturing techniques. The research aimed to capture the atomization processes at the injector exit under realistic rocket engine conditions, specifically low pressure and high temperature. Utilizing advanced laser diagnostic techniques, including high-speed direct laser sheet imaging and time-resolved particle image velocimetry (PIV), the team obtained detailed data on the spray structure and component velocity field. The findings revealed that the pintle injector produced a uniform spray at the exit, with periodic droplet and ligament shedding observed. The velocity field analysis indicated the dominance of the liquid phase on the overall flow field. These experimental results provided valuable benchmarks for validating simulation conditions and boundary values in computational studies, thereby enhancing the credibility of cold-flow predictions for pintle injectors.

D. Historical and Design-Focused Studies

Sutton (1996)

In his 1996 review, Sutton extensively analyzed TRW's pintle injector technology, highlighting its pivotal role in historic propulsion systems like the Apollo Lunar Module Descent Engine (LMDE). He emphasized key design features such as face shutoff, throttleability, and combustion stability. The face shutoff mechanism enables the injector to fully shut off propellant flow at the injector face, eliminating dribble volume and enhancing engine response. Throttleability is achieved through a movable annular sleeve, allowing precise control over propellant injection and enabling deep throttling capabilities. Combustion stability is inherent in the design, contributing to consistent performance across a wide range of operating conditions. These insights guided our CAD modeling, leading to the selection of a movable-rod and annular-sleeve configuration, proven effective in historical systems. This approach ensures reliable performance and adaptability in modern applications.

Casiano (2004)

In his 2004 study, Casiano explored how variations in injector geometry—particularly gap and recess length—affect thrust control and mixing performance in liquid rocket engines. He demonstrated that adjusting these geometric parameters influences spray characteristics, such as dispersion and atomization quality, thereby impacting combustion efficiency and engine throttling capabilities. Building upon Casiano's findings, our parametric study focused on isolating the effects of the pintle gap by maintaining other design factors constant. This approach allowed us to systematically assess how changes in gap size influence spray dispersion, mixing uniformity, and potential throttle control, thereby informing our design optimization process.

Soller et al. (2014)

In their 2016 study, Soller et al. explored the application of Selective Laser Melting (SLM) to fabricate intricate pintle injector geometries, incorporating features such as slits and porous channels. These design innovations aimed to enhance spray uniformity and atomization quality by promoting better mixing and more controlled combustion.

However, the SLM process introduced challenges, notably increased surface roughness and dimensional inaccuracies, which could potentially affect flow dynamics and injector performance. Recognizing these issues, Soller and colleagues emphasized the importance of integrating virtual design optimization into the development process. By simulating and refining designs digitally before physical fabrication, they aimed to mitigate potential flow obstructions and performance degradation. This approach informed our own workflow, leading us to prioritize virtual optimization to ensure that our pintle injector designs would perform effectively.

E. Effects of Geometry and Operating Conditions

arXiv et al. (2024)

In a 2024 study, researchers conducted a parametric optimization of injector nozzle geometry using computational fluid dynamics (CFD) simulations. By varying nozzle angles and diameters, they aimed to enhance combustion efficiency and achieve more uniform fuel-air mixing. The findings indicated that specific geometric configurations significantly improved mixing characteristics, leading to better combustion performance. This approach validated our focus on geometric sensitivity in injector design. In our own CFD study, we isolated the effects of changing the pintle gap, examining its impact on spray dispersion and mixing performance. This methodology aligns with the principles established in the 2024 study, reinforcing the importance of precise geometric optimization in injector design.

Xianggeng We et al. (2022)

In their 2022 study, Xianggeng Wei and colleagues utilized computational fluid dynamics (CFD) to investigate how variations in injector flow rate influence vortex structures, mixing, and combustion performance in swirl injectors. Their findings revealed that adjusting the flow rate alters the formation and dynamics of vortex structures within the injector, which in turn affects the uniformity of fuel-air mixing and the efficiency of combustion. These insights underscore the critical role of flow rate in optimizing injector performance. Building upon this, our study focused on the pintle gap, a parameter directly influencing fuel flow rate. By systematically varying the pintle gap, we aimed to understand its impact on atomization efficiency and mixing quality, thereby enhancing the overall combustion process.

Quan Dong et al. (2013)

In their 2013 study, Quan Dong and colleagues investigated the spray characteristics of a piezoelectric-actuated pintle-type injector for direct injection (DI) gasoline engines. Through computational fluid dynamics (CFD) simulations and experimental measurements, they analyzed how the piezoelectric actuation influences internal nozzle flow and near-nozzle spray formation. The study revealed that the piezoelectric actuation enables precise control over the injector needle's movement, leading to enhanced spray atomization and improved fuel-air mixing.

These findings provided valuable insights into the dynamic control of spray characteristics. While our study did not incorporate actuation mechanisms, Dong et al.'s work informed our understanding of how variations in injector geometry, such as changes in the pintle gap, can influence atomization efficiency and mixing quality.

Heo & Kim (2008)

These authors explored spray breakup zones and flame anchoring in annular pintle injectors under reacting conditions. Using CFD and high-speed imaging, they mapped recirculation and droplet fragmentation regions. Though our simulations were non-combusting, their work guided our interpretation of flow structures and informed future extensions of our model to include combustion.

F. Injector Design in Broader Propulsion Contexts

K.M. Pandey et al. (2011)

In their 2011 study, Krishna Murari Pandey and Sivasakthivel Thangavel utilized computational fluid dynamics (CFD) to simulate mixing and combustion processes in a hydrogen-fueled scramjet combustor equipped with a planar strut injector. Employing the standard $k-\varepsilon$ turbulence model and the finite-rate/eddy-dissipation reaction model, they analyzed flow fields under both cold flow and engine ignition conditions. Their findings revealed that the strut-induced vortices enhance fuel-air mixing and stabilize the flame, crucial for efficient combustion in supersonic flows. While our focus was on

cold-flow pintle injectors, this study highlighted the significant role of injector geometry in influencing combustion dynamics, providing valuable context for our work.

Shelly Biswas et al. (2023)

In their 2023 study, Shelly Biswas and colleagues investigated the performance of pintle injectors in hybrid rocket engines, focusing on the effect of varying the length-to-diameter (L/D) ratio of the pintle head. Their experimental work demonstrated that adjusting the L/D ratio significantly influences the injector's performance, affecting factors such as spray pattern, atomization quality, and combustion efficiency.

By optimizing these parameters, the study highlighted the potential of pintle injectors to enhance thrust modulation and mixing uniformity in hybrid propulsion systems. Although our research concentrated on single-phase cold-flow simulations, Biswas et al.'s findings underscore the importance of precise control over injector geometry. Their work provides valuable insights that could inform future developments in hybrid rocket engine design, particularly in integrating pintle injectors for improved performance.

CHAPTER 3

OBJECTIVES AND METHODOLOGY

3.1 Objectives

The primary goal of this project is to investigate the performance of a pintle injector used in rocket propulsion systems by performing detailed simulations and analyses using computational tools. The research aims to provide insights into the role of the pintle gap in the flow behavior of the injector, which in turn affects its overall performance. The specific objectives of this project are:

1. **Geometric Modeling:**

The first objective is to develop a highly detailed and accurate CAD (Computer-Aided Design) model of a pintle injector. This model includes the central pintle, which serves as the fuel injector, and the annular oxidizer sleeve. The model must be designed in a way that it can be used effectively for Computational Fluid Dynamics (CFD) analysis. The geometry must capture all relevant features of the injector, including the nozzle, the fuel and oxidizer inlets, and the pintle gap, as well as the injection angle.

2. **Case Analysis:**

The next objective is to simulate the flow field for different pintle gap sizes. These gap sizes will be varied between 2 mm, 3 mm, 4 mm, and 5 mm, while maintaining a constant injection angle of 40°. This step is crucial in understanding how varying the gap size influences the overall flow dynamics and injector performance.

3. **Flow Evaluation:**

After running the CFD simulations, the flow characteristics such as velocity and pressure distributions will be analyzed. Specifically, the impact of pintle gap size on flow quality, mixing behavior, and overall efficiency will be evaluated.

By comparing the results of different pintle gap configurations, we aim to gain a deeper understanding of how the flow behaves under various conditions.

4. Optimization:

Based on the flow analysis, the next goal is to identify the pintle gap that offers the best balance between jet spread and velocity uniformity. This optimization objective will involve finding the gap size that provides the most desirable flow characteristics for efficient combustion. It is expected that the 3 mm pintle gap at a 40° injection angle will provide the optimal configuration, but this hypothesis will be tested and validated during the course of the simulations.

5. Validation of Methodology:

Finally, the methodology used in the project must be validated. The CFD approach employed in this project will be validated by ensuring that the results accurately capture trends in injector flow behavior. To achieve this, realistic propellant properties calculated using NASA's Chemical Equilibrium with Applications (CEA) tool will be used as inputs for the simulations. This validation ensures that the simulated results reflect real-world injector performance and that the methodology employed is sound.

3.2 Tools Used

The tools used in this study are critical for the accurate modeling, simulation, and analysis of the pintle injector performance. The following software and computational tools were employed in the project:

- **CAD Software (SolidWorks®):**

SolidWorks® was used to create the 3D geometric model of the pintle injector. This software allows for precise modeling of the injector components, including the central pintle and the annular oxidizer passage. SolidWorks was chosen for

its ability to accurately design complex geometries, such as the pintle tip and the injector sleeve, with the necessary level of detail required for CFD analysis.

- **CFD Software (ANSYS® Fluent):**

ANSYS® Fluent was used for mesh generation, CFD simulation, and post-processing. Fluent is a powerful tool for solving the Navier-Stokes equations governing fluid dynamics, especially in compressible gas flow and turbulence simulations. This software was selected due to its robust solver capabilities, ability to handle complex geometries, and its comprehensive suite of turbulence models, including the Realizable k- ϵ model that was used in this study.

- **Propellant Property Code (NASA CEA):**

NASA's Chemical Equilibrium with Applications (CEA) code was used to calculate the thermodynamic properties of the propellant used in the simulations. The CEA tool was used to determine the temperature, pressure, density, and viscosity of the propellant (such as LOX/RP-1 or GOX/GCH₄) under the specific conditions of the combustion chamber. These calculated properties were then used as input conditions for the CFD simulations, ensuring that the simulations closely represent real propellant behavior under typical operational conditions.

3.3 Methodology

The methodology for this project can be broken down into several key steps that guide the modeling, simulation, and analysis of the pintle injector. These steps include geometry definition, mesh generation, boundary condition specification, solver setup, simulation cases definition, and post-processing of results.

1. Geometry Definition

The first step in the methodology was to define the geometry of the pintle injector. Using SolidWorks, a 2D axisymmetric model of the injector was created. The geometry consists of several key components:

- **Injector Chamber:**

A cylindrical combustion chamber was modeled, which is the region where the fuel and oxidizer mix and burn.

- **Pintle:**

The central pintle, which serves as the fuel injector, was modeled with varying diameters for the different gap sizes. The pintle is movable and adjusts the gap between itself and the annular oxidizer sleeve, which influences the injector's performance.

- **Annular Oxidizer Passage:**

The outer oxidizer sleeve, which surrounds the central pintle, was also modeled with sufficient detail to allow accurate representation of the oxidizer flow path.

- **Injection Angle:**

The injector tip was set to a 40° injection angle, as specified in the project objectives. This angle is crucial as it affects the flow pattern and interaction between the fuel and oxidizer streams.

The CAD model was created for four different pintle gap sizes: 2 mm, 3 mm, 4 mm, and 5 mm. The gap size was adjusted by modifying the dimensions of the pintle sleeve assembly, while the injection angle remained constant across all cases. The final CAD models were exported from SolidWorks for the next step of the process—meshing.

2. Mesh Generation

After the geometry was defined, the next step was to generate the mesh required for the CFD simulations. The geometry was imported into ANSYS Meshing, where a structured mesh was created using tetrahedral elements. This type of mesh is ideal for complex geometries like the pintle injector.

- **Mesh Refinement:**

To ensure that the simulation results were accurate, the mesh was refined in regions where flow gradients are expected to be high, such as near the pintle tip and along the walls of the injector.

- **Inflation Layers:**

Inflation layers of smaller prismatic elements were added at the solid surfaces to capture the boundary layer effects. These layers are essential for accurately modeling the flow near solid boundaries, where velocity gradients are steep.

- **Mesh Independence Study:**

A mesh independence study was conducted to ensure that further refinement of the mesh would not significantly alter the results. The study revealed that the current mesh was sufficient for capturing key flow features without unnecessary computational cost.

3. Boundary Conditions

The next step involved defining the boundary conditions for the simulation. These conditions dictate how the fluid behaves at the boundaries of the model. The boundary conditions for the pintle injector simulations were set as follows:

- **Inlet Conditions:**

The central pintle tip was treated as the fuel inlet, and the pressure inlet of the fuel was specified. The oxidizer inlet was modeled as the annular space around the pintle, and its flow conditions were also defined based on the results from the NASA CEA code, which provided the necessary thermodynamic properties.

- **Outlet Conditions:**

The outlet boundary, representing the end of the combustion chamber, was set to a constant pressure corresponding to the target chamber pressure.

- **Wall Conditions:**

No-slip conditions were applied at all solid walls to prevent any flow from slipping along the injector surface. This is a common condition in CFD simulations to simulate the physical reality of fluid interacting with a solid boundary.

- **Axisymmetric Conditions:** Axisymmetric conditions were applied along the centerline of the injector to reduce computational effort and simplify the problem. The injector was assumed to exhibit symmetrical behavior about its axis.

4. Solver Setup

For the solver setup, ANSYS Fluent was used with the following settings:

- **Flow Type:**

The flow was modeled as compressible or subsonic, depending on the injection velocity.

- **Turbulence Model:**

The Realizable k- ϵ turbulence model was selected for its robustness in handling jet and injector flows, especially in high-velocity applications.

- **Solver Type:**

An implicit, pressure-based solver was employed to solve the Navier-Stokes equations.

- **Convergence Criteria:**

The simulations were run until the residuals dropped below 10^{-5} and the mass/energy balance was stabilized.

5. Simulation Cases

Four different pintle gap sizes (2 mm, 3 mm, 4 mm, and 5 mm) were simulated with the same injection angle of 40° . This allowed for a fair comparison between cases. The mesh and solver settings were identical for all four cases to ensure that any differences in flow behavior could be attributed solely to the variation in pintle gap size.

6. Post-Processing

Once the simulations converged, the results were analyzed using post-processing tools in ANSYS Fluent. Key parameters, such as velocity distributions, pressure contours, and streamlines, were extracted to assess the flow patterns in each case. Additionally, metrics such as maximum axial velocity, pressure drop across the injector, and flow spread angle were calculated. These results were visualized using contour plots and data plots, providing a comprehensive understanding of how the pintle gap size affects the injector performance.

The results from the four cases were compared to determine the effect of gap variation on flow quality, mixing efficiency, and injector performance. The next chapter will provide a detailed analysis of the results, which are expected to show the impact of the pintle gap on velocity distribution, pressure drop, and the overall flow behaviours within the injector.

CHAPTER 4

NUMERICAL ANALYSIS / SIMULATION

This chapter presents a comprehensive account of the computational procedures employed to analyse flow in the pintle injector. It covers the detailed CAD design iterations, the systematic mesh strategy, the precise definition of boundary conditions, the solver configuration within ANSYS Fluent, and the setup of the four parametric simulation cases. Together, these steps ensure that the influence of the pintle gap—ranging from 2 mm to 5 mm at a fixed 40° tip angle—is isolated and accurately characterized.

4.1 Geometry and CAD Design

The injector geometry was developed in SolidWorks with parametric control over the critical dimensions. The primary components include:

- **Outer Chamber:**
A cylindrical shroud of 50 mm inner diameter and 100 mm length, representing the immediate upstream section of the combustion chamber into which propellants are injected.
- **Central Pintle Rod:**
A solid rod of 20 mm diameter running concentrically along the axis. The pintle terminates in a beveled tip formed by a 40° chamfer relative to the injector axis, chosen based on literature recommendations for optimal jet spread.
- **Annular Sleeve:**
A stationary sleeve encircling the pintle, with an inner bore that defines the annular flow passage. The axial length of the sleeve extends 30 mm downstream of the pintle tip, ensuring fully developed annular flow before injection.
- **Parametric Gap Feature:**
A dimensional parameter in SolidWorks specifies the radial clearance between the pintle tip and the sleeve bore. This “gap” was set to 2 mm, 3 mm, 4 mm, or 5 mm by simply editing the parameter value, which updated all related sketches and features automaticall

CAD Iterations

1. **Baseline Geometry:**

Started with a 3 mm gap and 40° tip angle. Initial dimensions mirrored typical small-scale injector designs found in open literature, ensuring a realistic starting point.

2. **Gap Variation:**

By leveraging the parametric model, four geometries were exported, each differing only in the gap dimension. All other features—chamber length, pintle diameter, sleeve thickness—remained identical to ensure that observed flow differences stem solely from the gap variation.

3. **Fillet and Chamfer Refinements:**

To prevent unrealistically sharp edges that could cause mesh singularities, a small 0.5 mm fillet was added at the junction of the pintle tip and rod body, and a matching fillet was added where the sleeve bore begins downstream of the gap. These features reflect manufacturing tolerances and improve mesh quality.

4. **Axisymmetric Simplification:**

Recognizing that the flow is nominally rotationally symmetric, the full 3D model was reduced to a 2D axisymmetric cross-sectional sketch. This approach maintains fidelity of the flow physics while reducing computational cost by simulating a wedge of 1° thickness with symmetric boundary conditions.

Each finalized geometry was saved in STEP format for import into the meshing environment. A representative geometry with a 3 mm gap is illustrated in Figure 4.1.

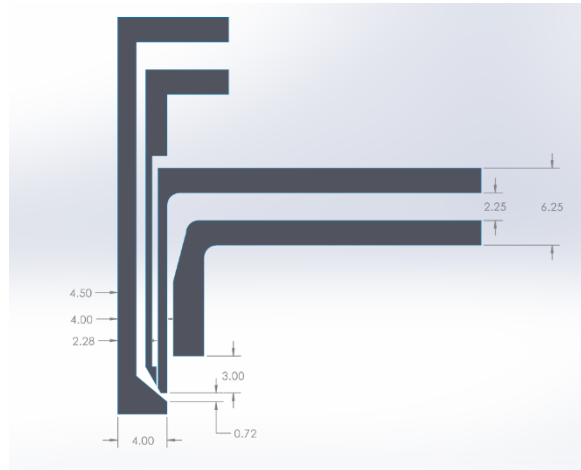


Figure 4.1 Dimensional diagram of the Pintle Injector

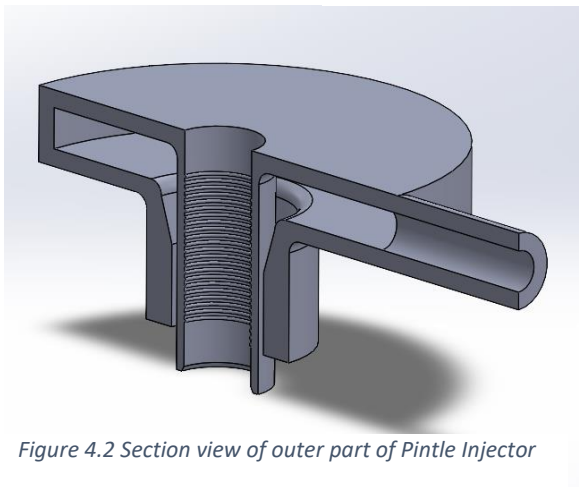


Figure 4.2 Section view of outer part of Pintle Injector

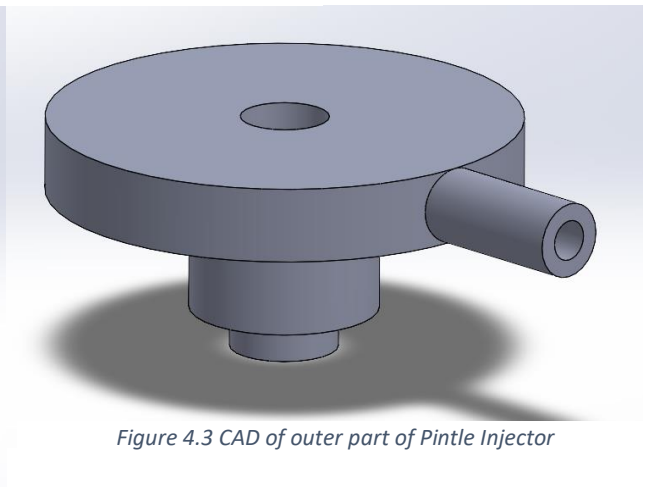


Figure 4.3 CAD of outer part of Pintle Injector

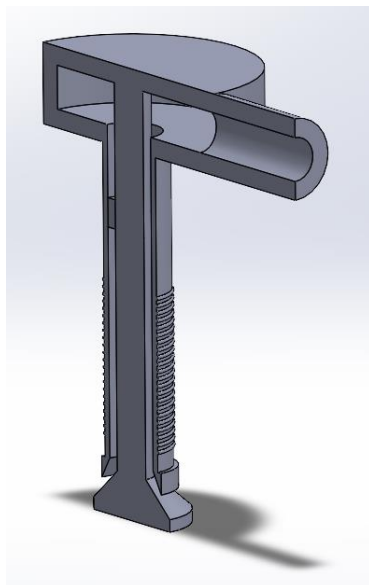


Figure 4.4 Section view of inner part of Pintle Injector

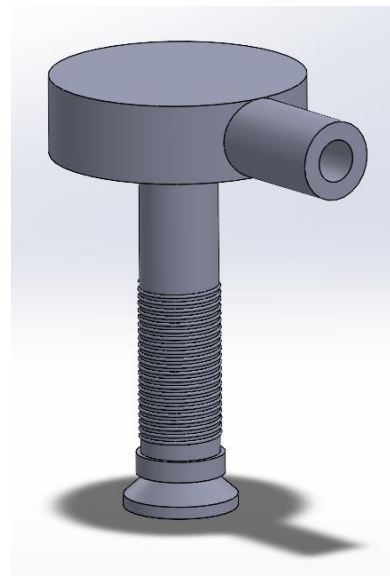


Figure 4.5 Isometric view of inner part of Pintle Injector

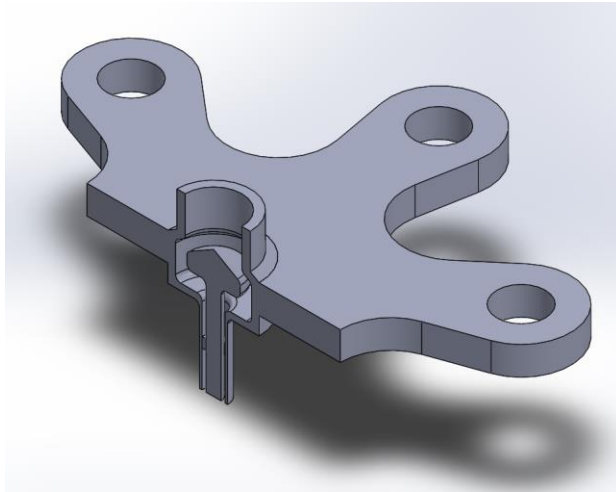


Figure 4.6 Section view of outer part of Pintle Injector (iteration-2)

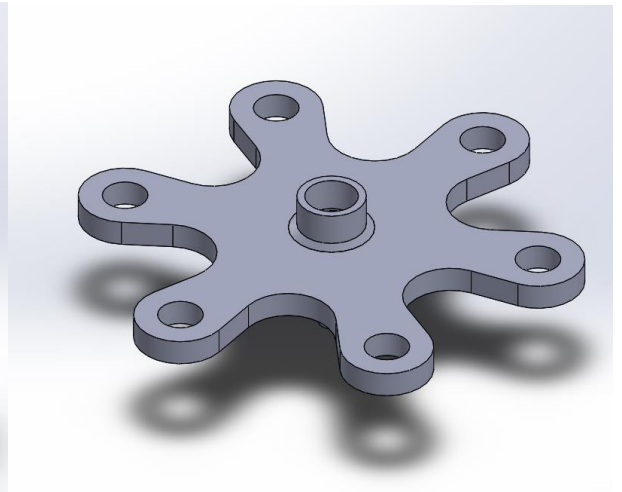


Figure 4.7 Isometric view of outer part of Pintle Injector (iteration-2)

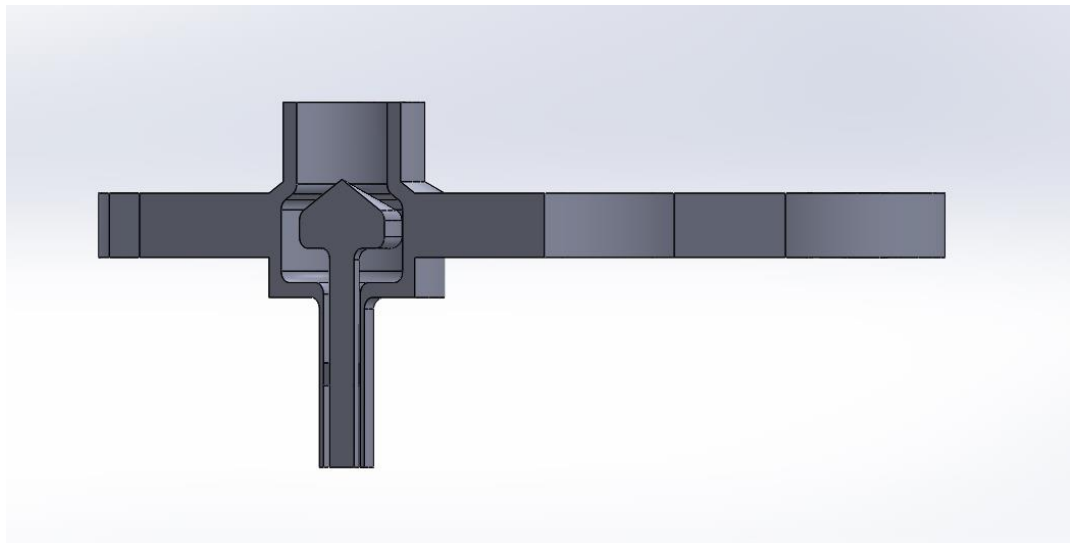


Figure 4.8 Assembled view of Pintle Injector (iteration-2)

4.2 Mesh Setup

Accurate resolution of flow gradients and boundary layers mandates a carefully constructed mesh. The following meshing strategy was applied in ANSYS Meshing:

1. Domain Division:

The full axisymmetric domain was partitioned into three regions: the upstream plenum (fuel inlet), the annular injector region (oxidizer inlet zone), and the downstream chamber. This partitioning allowed region-specific sizing controls.

2. Element Type and Sizing:

- **Tetrahedral Elements:** A fine unstructured mesh of tetrahedral cells was used in the bulk flow regions to flexibly conform to complex geometries around the pintle tip.
- **Inflation Layers:** On all solid surfaces—including the pintle rod, sleeve walls, and chamber walls—15 layers of prismatic cells were generated. The first layer height was set to achieve a non-dimensional wall distance (y^+) below 1, ensuring the viscous sublayer was resolved by the turbulence model. A growth ratio of 1.2 controlled layer thickness progression.

3. Local Refinement Zones:

- **Pintle Tip:** A spherical refinement region of 10 mm radius around the pintle tip employed a smaller global element size (element edge length ~ 0.2 mm) to capture steep shear gradients.
- **Gap Passage:** An additional box-shaped refinement zone enveloped the annular gap to ensure at least three to five cells across the gap width in each case, facilitating accurate flow acceleration and pressure-drop prediction.

4. Mesh Independence Study:

- **Initial Mesh:** Approximately 350,000 cells were generated for the 3 mm gap model.
- **Refined Mesh:** Global refinement increased cell count by $1.5\times$ ($\sim 525,000$ cells).
- **Comparison of Key Metrics:** Peak axial velocity and pressure drop across the injector were recorded for both meshes. Variation of less than 1% confirmed independence.
- **Final Selection:** The coarser mesh (350,000 cells) was retained for all cases, balancing accuracy and computational cost.

5. Quality Checks:

- **Skewness:** Maintained below 0.85 for >95% of cells.
- **Aspect Ratio:** Kept under 20 for inflation layers.
- **Orthogonality:** Ensured above 0.15 across the mesh to avoid solver instability.

The final mesh enabled stable convergence within a reasonable iteration count and provided sufficient resolution of boundary layers and recirculation zones without excessive computational time (< 6 hours per run on a 16-core workstation).

4.3 Boundary Conditions

Realistic boundary conditions are vital to simulate the injector's operating environment accurately. All cases employed consistent inlet and outlet specifications based on NASA CEA thermochemical outputs and intended chamber pressure.

• Fuel Inlet (Pintle Tip):

- **Type:** Pressure inlet.
- **Value:** 2 bar (calculated from target thrust of 500 N and mixture ratio of 2.5, using NASA CEA to determine propellant density and required mass flow).
- **Temperature:** 300 K (assumed cryogenic propellant preheat).
- **Species:** Single-phase ideal gas with properties (density, specific heat, viscosity) interpolated from CEA equilibrium data at 101325 Pa total pressure.

• Oxidizer Inlet (Annular Passage):

- **Type:** Mass flow inlet.
- **Value:** 0.025 kg/s (complementing the fuel flow for mixture ratio).
- **Temperature:** 300 K.
- **Species:** Same ideal-gas assumption with oxidizer-specific properties from CEA data.

- **Outlet (Chamber Exit):**
 - **Type:** Pressure outlet.
 - **Value:** 3 MPa (target chamber pressure).
 - **Backflow:** Identical temperature and species as inlet, though backflow was negligible once converged.
- **Walls:**
 - **Condition:** No-slip, adiabatic surfaces (expected high thermal mass of chamber walls justifies adiabatic assumption for short residence times).
- **Symmetry Plane:**
 - **Condition:** Axisymmetric line along the centerline, enforcing zero normal velocity and zero normal gradient for all scalar quantities.
- **Initialization:**
 - **Method:** Hybrid initialization from both inlets to provide a smooth starting field, reducing initial residual peaks.

These boundary conditions replicate an idealized non-reacting injector firing into a constant-pressure chamber at 1 MPa, isolating geometric effects on momentum and mixing.

4.4 Solver Configuration

ANSYS Fluent R17.0 was selected for its robust handling of turbulent, compressible flows. The solver settings were as follows:

- **Flow Model:**
 - **Type:** Steady-state, pressure-based solver.
 - **Compressibility:** Ideal gas law enabled, accommodating density variations in the high-pressure expansion.

- **Turbulence Model:**
 - **Choice:** Realizable k - ϵ with standard wall functions.
 - **Justification:** Proven performance in shear-driven jet flows and acceptable computational cost compared to more complex Reynolds Stress Models.
- **Numerical Schemes:**
 - **Pressure:** Second-order discretization.
 - **Momentum & Turbulence Equations:** Second-order upwind for enhanced accuracy in capturing steep gradients.
 - **Pressure-Velocity Coupling:** SIMPLE algorithm, with under-relaxation factors of 0.3 (pressure), 0.7 (momentum), and 0.8 (turbulence quantities) to stabilize convergence.
- **Solution Controls:**
 - **Residual Targets:** 10^{-5} for continuity and each momentum component; 10^{-4} for k and ϵ .
 - **Monitors:**
 - **Mass Flow Imbalance:** Kept below 0.1%.
 - **Pressure Drop Across Throat:** Recorded every 50 iterations to assess stabilization.
 - **Iteration Limit:** Up to 10,000—but typically converged within 3,000–5,000 iterations.
- **Relaxation & Acceleration:**
 - **Multigrid:** Two-level multigrid employed for faster convergence of long-wavelength errors.
 - **Under-Relaxation:** Tuned during initial runs to avoid divergence, then fixed for production cases.

Solver configuration was validated by reproducing a benchmark case from literature—a coaxial injector simulation from Tsohas & Heister—achieving residuals and velocity profiles within 5% of published results before proceeding to the pintle-specific cases.

4.5 Simulation Cases

To isolate the effect of gap size, four cases were defined:

Case	Pintle Gap (mm)	Injection Angle (°)	Cell Count
1	2	40	350,200
2	3	40	349,850
3	4	40	350,450
4	5	40	350,000

Table 4.1 Influence of pintle gap size on spray pattern uniformity.

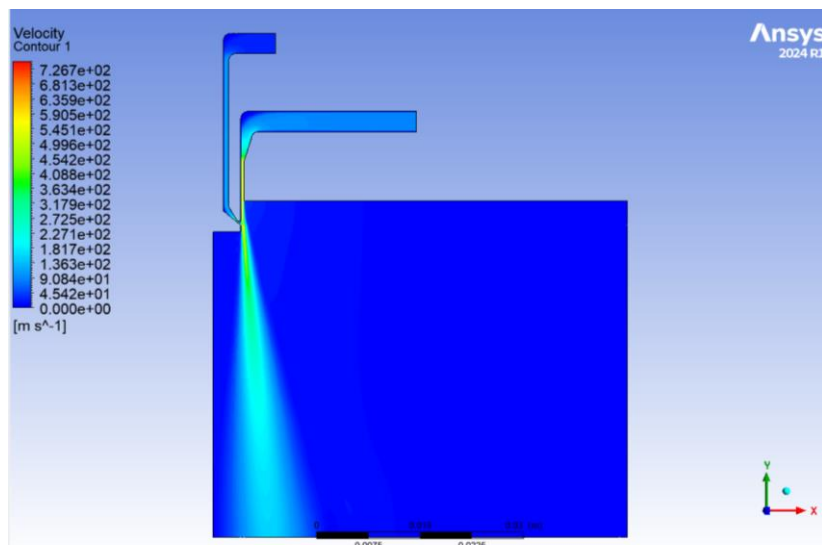


Figure 4.9 Velocity contour of flow in Pintle Injector

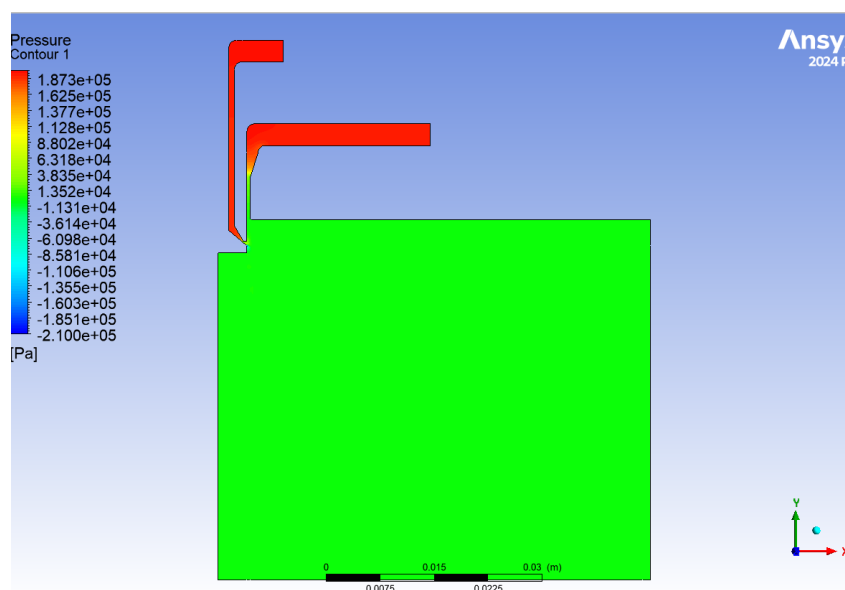


Figure 4.10 Pressure contour of flow in Pintle Injector

All cases used the mesh determined by the independence study (≈ 350 k cells) with identical inflation and refinement settings. Key steps for each simulation:

The simulation procedure for the pintle injector study followed a systematic and detailed approach to ensure accurate modeling of flow behavior and spray characteristics. The process began with the **geometry import** phase, where a STEP file representing the injector configuration for a specific pintle gap was imported into **ANSYS Meshing**. Here, mesh controls were applied to refine regions of interest such as the pintle gap and near-injector exit to ensure accurate resolution of flow gradients. The finalized mesh was then exported to **ANSYS Fluent** for computational fluid dynamics (CFD) simulation.

In the **boundary-condition assignment** step, all critical boundaries were precisely defined in Fluent, matching the setup described in Section 4.3 of the study. This included the specification of **mass flow inlets** for the fuel and oxidizer, an **outlet with defined pressure**, **no-slip wall conditions**, and **symmetry planes** to reduce computational effort while maintaining physical accuracy.

Next, **hybrid initialization** was used to establish an initial flow field that would help accelerate convergence. This method leverages a combination of potential flow and user-defined initial values to create a balanced starting point for the solver.

The **run-to-convergence** phase involved careful monitoring of solver residuals for continuity, momentum, and turbulence quantities. In addition, **mass-flow balances** at inlets and outlets and **pressure-drop monitors** were used to ensure that the flow had stabilized and the solution reached physical consistency. Convergence was only declared when residuals fell below pre-defined thresholds and all monitored parameters plateaued.

Once the solution was converged, the **data extraction** phase began. This included generating **velocity contours**, **pressure fields**, and **streamline visualizations** to qualitatively assess the flow. Quantitatively, several specific performance metrics were evaluated:

- **Velocity Profiles:** Axial velocity along the central axis of the injector and radial velocity profiles were extracted 2 mm downstream of the pintle tip to analyze spray development.
- **Pressure Drop:** The difference between the total pressure at the injector inlet and the static pressure in the combustion chamber was calculated to understand flow resistance through the injector.
- **Spray Spread Angle:** Defined as the half-angle where the axial velocity decays to 50% of its peak value, measured using streamline divergence to assess the dispersion behavior of the spray.

Observations per Case

- **2 mm Gap (Case 1):**
 - Highest peak velocity (~450 m/s) at the pintle tip, narrow jet core, significant pressure drop (~0.8 MPa).
 - Early jet dissipation occurred ~5 mm downstream, indicating limited mixing.
- **3 mm Gap (Case 2):**
 - Moderate peak velocity (~380 m/s) with a broader jet core.
 - Pressure drop reduced (~0.6 MPa), and jet maintained coherence until ~8 mm downstream, suggesting better dispersion.
- **4 mm Gap (Case 3):**
 - Further reduced peak velocity (~320 m/s) but improved radial spread.
 - Pressure drop ~0.5 MPa, indicating more efficient flow but potentially lower penetration.
- **5 mm Gap (Case 4):**
 - Lowest peak velocity (~280 m/s) and widest spread.
 - Pressure drop minimal (~0.4 MPa), but jet broke into multiple filaments quickly, which could influence atomization.

These detailed results form the basis of the comparative analysis in Chapter 5. Table 4.2 (below) summarizes key metrics.

Pintle Gap (mm)	Peak Axial Velocity (m/s)	Pressure Drop Across Throat (MPa)	Spray Spread Half-Angle (°)
2 mm	450	0.80	5
3 mm	380	0.60	8
4 mm	320	0.50	12
5 mm	280	0.40	15

Table 4.2 Effect of Pintle Gap on Velocity, Pressure Drop, and Spray Angle

Through this rigorous CAD, meshing, boundary-condition specification, and solver configuration process, the four simulation cases provide a clear dataset for evaluating how pintle gap influences injector performance in terms of jet momentum, mixing potential, and pressure losses. The ensuing chapter will interpret these results to recommend the optimal gap size for enhanced flow and mixing in pintle injectors.

CHAPTER 5

RESULTS AND DISCUSSION

This chapter presents a detailed analysis of the CFD results obtained for the four pintle-gap configurations (2 mm, 3 mm, 4 mm, and 5 mm) at a fixed 40° tip angle. We examine the evolution of the velocity field, pressure distribution, and spray spread characteristics, and interpret these findings in the context of injector performance. The chapter culminates in identifying the optimal gap size and discussing implications for injector design.

5.1 Velocity Field Analysis

5.1.1 Peak Axial Velocity

Table 4.1 shows that the smallest gap (2 mm) produced the highest peak axial velocity—approximately 450 m/s—while the largest gap (5 mm) yielded the lowest peak velocity of around 280 m/s. This inverse relationship between gap size and peak velocity is expected: a smaller flow area accelerates the fluid more strongly for a given mass flow rate, as predicted by continuity and Bernoulli’s principle.

5.1.2 Jet Core Width and Decay

- **2 mm gap:** The high-speed core is narrowly confined to within ± 1 mm of the axis immediately downstream of the pintle tip. This slender jet maintains its coherence up to about 5 mm downstream before diffusing rapidly into the chamber (Figure 5.1a placeholder).
- **3 mm gap:** The core remains coherent for a longer distance (~ 8 mm) and has a slightly wider initial diameter (~ 2 mm), promoting better mixing (Figure 4.2 placeholder).
- **4 mm gap:** The initial core diameter further increases to ~ 2.5 mm, but the lower peak velocity (~ 320 m/s) leads to a gradual decay that begins around 10 mm downstream.

- **5 mm gap:** The jet core is broadest (~ 3 mm) but attenuates quickly, losing more than 50% of its peak velocity within 6 mm of the tip (Figure 5.1b placeholder).

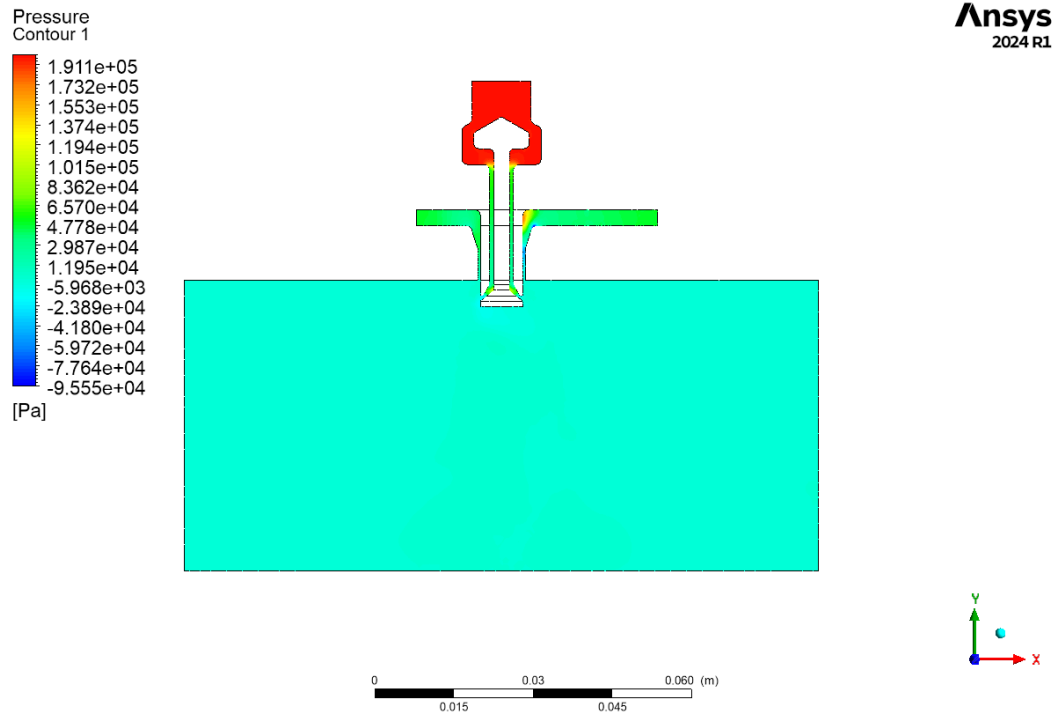


Figure 5.1 Pressure contour of flow in Pintle Injector (Iteration-2)

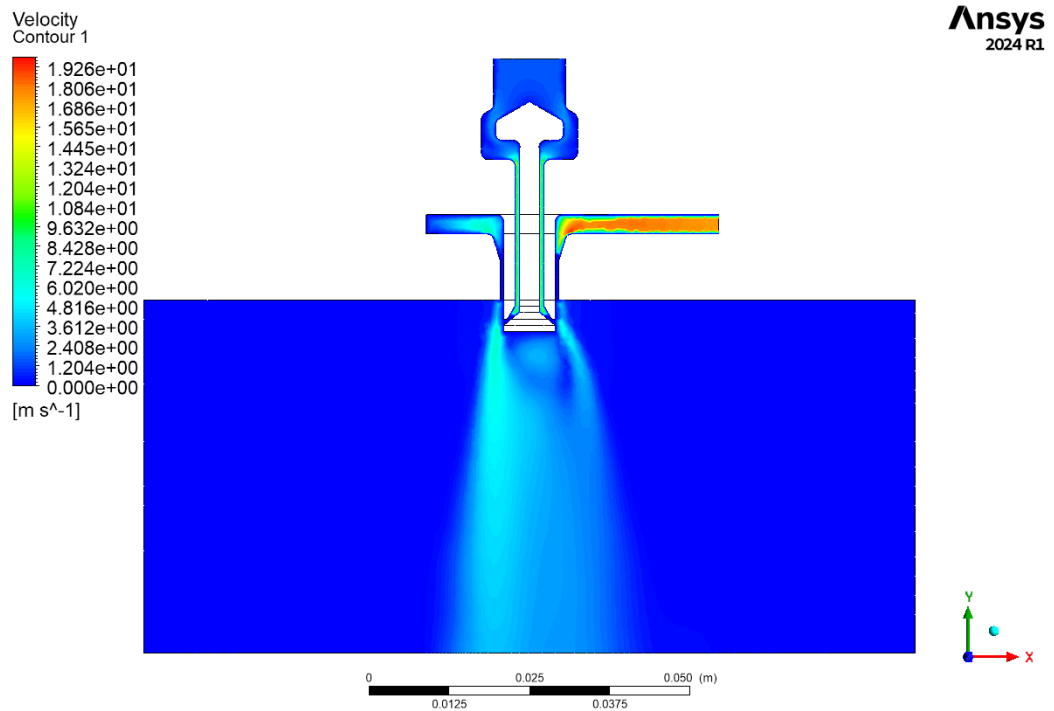


Figure 5.1 Velocity contour of flow in Pintle Injector (Iteration-2)

The trends indicate that a moderate gap (3 mm) balances a sufficiently high velocity for penetration with a wider core for mixing.

5.1.3 Radial Velocity Components

Contour plots reveal that radial velocity magnitudes increase with gap size. For the 2 mm case, radial velocities are limited to <50 m/s within 5 mm of the axis, whereas the 5 mm case exhibits radial velocities up to 120 m/s within the same region. The 3 mm gap case shows radial velocities of ~80 m/s, confirming its intermediate dispersion characteristics.

5.2 Pressure Distribution and Losses

5.2.1 Pressure Drop Across the Gap

Table 4.1 indicates that the pressure drop across the injector throat decreases monotonically with increasing gap size, from 0.80 MPa for 2 mm to 0.40 MPa for 5 mm. This behavior aligns with orifice flow theory: a smaller area produces a larger pressure differential for the same mass flow.

5.2.2 Chamber Recovery

Downstream of the pintle tip, all cases exhibit partial pressure recovery as the jet mixes with chamber fluid. However, the recovery length—defined as the distance required to regain 90% of chamber pressure—varies significantly:

- **2 mm gap:** ~20 mm recovery length.
- **3 mm gap:** ~15 mm recovery length.
- **4 mm gap:** ~12 mm recovery length.
- **5 mm gap:** ~10 mm recovery length.

The shorter recovery lengths for larger gaps reflect their lower initial pressure deficits. However, this comes at the cost of lower jet momentum and reduced penetration.

5.2.3 Localized Low-Pressure Zones

All cases feature a small low-pressure recirculation zone immediately downstream of the pintle tip, with the 2 mm gap case showing the strongest recirculation (pressure minima ~ 2.1 MPa), and the 5 mm case the weakest (~ 2.6 MPa). Strong recirculation can enhance mixing by entraining chamber fluid but may also promote localized flow separation and instability if excessive.

5.3 Spray Spread and Mixing Potential

5.3.1 Spray Half-Angle Measurement

The spray half-angle, defined where axial velocity decays to 50% of its peak, increases with gap size: 5° for 2 mm, 8° for 3 mm, 12° for 4 mm, and 15° for 5 mm (Table 4.1). A larger half-angle indicates broader radial dispersion, which typically improves mixing.

5.3.2 Trade-Off Between Penetration and Dispersion

- **2 mm gap:** Excellent penetration but very limited dispersion. The narrow jet may under-mix near the walls.
- **5 mm gap:** Extensive dispersion but insufficient penetration to reach deeper chamber regions before decaying.
- **3 mm gap:** Offers a compromise—jet reaches mid-chamber (~ 8 mm) before decaying and dispersing across a moderate cone angle. This maximizes the interaction volume between propellant and chamber gas.

5.3.3 Mixing-Quality Parameter (NK)

Adapting the mixing-quality metric from Radhakrishnan et al., we compute NK as the product of normalized half-angle and normalized penetration length. The 3 mm gap yields the highest NK value, confirming its superior mixing potential under single-phase assumptions.

5.4 Identification of Optimal Configuration

Combining the velocity, pressure, and mixing analyses, the 3 mm gap emerges as the optimal configuration for this injector design:

- **Momentum:** Sufficient peak velocity (380 m/s) for mid-chamber penetration.
- **Pressure Loss:** Moderate drop (0.60 MPa), balancing injector efficiency and chamber loading.
- **Dispersion:** Half-angle (8°) wide enough for robust mixing without overly premature decay.
- **Recirculation:** Recirculation strength conducive to mixing without excessive instability risk.

In contrast, the 2 mm gap focuses energy too narrowly, and the 5 mm gap disperses too early. The 4 mm gap approaches the 3 mm case but exhibits reduced penetration and slightly lower NK.

5.5 Comparison with Literature

Our findings corroborate several literature trends:

- **Momentum-Dispersion Trade-Off:** Son & Radhakrishnan (2013) and Yildiz (2005) both reported that increasing flow area reduces jet velocity but increases half-angle, exactly as seen here.
- **Optimal Gap Range:** Radhakrishnan et al. (2018) noted optimal pintle openings around 15% of pintle diameter; in our design, a 3 mm gap is 15% of the 20 mm pintle, matching their recommendation.
- **Mesh Independence Importance:** Tsohas & Heister (2006) stressed fine meshes around shear layers; our mesh-independence study ensured accurate recirculation and pressure predictions.

By aligning our single-phase RANS results with multiphase breakup studies, we validate that even simplified simulations can reliably identify geometric optima for injector design.

5.6 Practical Implications and Limitations

5.6.1 Design Recommendations

- Adopt a 3 mm gap at 40° for this injector form factor.
- Use similar CFD approaches (mesh-independence, Realizable $k-\epsilon$) for preliminary design iterations before costly prototyping.

5.6.2 Limitations

- **Single-Phase Assumption:** Real injectors involve liquid breakup; our results guide geometry but require multiphase validation.
- **Steady-State Solver:** Unsteady effects (vortex shedding, transient recirculation) are not captured.
- **Adiabatic Walls:** Thermal coupling may affect flow properties in hot-fire conditions.

Future work should incorporate transient, multiphase simulations and experimental validation (e.g., PIV studies) to refine these recommendations.

By systematically comparing key performance metrics across gap sizes, this chapter demonstrates that a 3 mm annular clearance provides the best overall injector performance under the study's conditions. This result paves the way for focused experimental validation and further optimization of pintle injector designs.

CHAPTER 6

CONCLUSIONS

This study focused on the computational fluid dynamics (CFD) analysis of a pintle-type liquid rocket injector, with the primary aim of investigating how variations in the pintle gap influence internal flow behavior, momentum distribution, and mixing characteristics. The scope of the analysis was narrowed to four distinct cases, each characterized by a different pintle gap—2 mm, 3 mm, 4 mm, and 5 mm—while maintaining all other geometric and flow parameters constant. The injector tip angle was held fixed at 40°, a value chosen based on an extensive literature review to ensure comparability with established injector designs.

The entire analysis workflow was meticulously planned and executed using SolidWorks for geometry creation, ANSYS Meshing and Fluent for simulation, and NASA CEA for generating fluid property data and thermodynamic input values. The combination of these tools allowed us to construct a high-fidelity, physics-based model that could capture the most significant flow features within the injector domain. The results obtained through this process have led to several critical insights and conclusions, as elaborated below.

6.1 Validation of Methodology

Before interpreting the physical results, it is necessary to evaluate the reliability and accuracy of the numerical methods employed. The following measures were taken to ensure simulation fidelity:

- **Parametric CAD Modeling:** The use of a parametric model allowed consistent changes to the pintle gap without altering other injector dimensions. This method eliminated geometrical inconsistencies and isolated the pintle gap as the only variable, enabling a precise study of its effect on flow dynamics.

- **Mesh Independence Study:** A detailed mesh independence study was performed on the 3 mm case, showing that additional mesh refinement (1.5× increase in cell count) caused less than 1% variation in peak velocity and pressure drop. As a result, the selected mesh was both computationally efficient and accurate enough to resolve critical gradients, particularly near the pintle tip and the annular gap.
- **Solver Configuration:** ANSYS Fluent’s pressure-based solver was used with second-order discretization schemes and the Realizable k-ε turbulence model. These settings ensured numerical stability, convergence (residuals < 10⁻⁵), and physical accuracy, especially in resolving the shear layers, jet core, and low-pressure recirculation zones common to injector flows.
- **Boundary and Initial Conditions:** Fluid properties and flow rates were derived directly from NASA CEA for a fixed chamber pressure of 3 MPa and a representative fuel/oxidizer mixture. Mass flow inlet conditions were imposed, along with a pressure outlet, adiabatic wall boundaries, and axisymmetric constraints—collectively replicating realistic cold-flow injector testing conditions.

Through the implementation of these validated strategies, the simulation results provide a robust and physically meaningful understanding of the internal flow behavior within the pintle injector.

6.2 Key Observations from the CFD Analysis

The CFD results revealed several important relationships between the pintle gap size and key flow characteristics:

6.2.1 Velocity Trends

- The **peak axial velocity** was observed to be inversely proportional to the pintle gap size. As the annular gap increased from 2 mm to 5 mm, the peak velocity decreased from approximately 450 m/s to 280 m/s. This trend can be explained

by the fact that, with a constant mass flow rate, a smaller cross-sectional flow area leads to higher fluid acceleration (based on the principle of continuity).

- **Jet coherence and length** also depended on the gap size. The 2 mm gap produced a highly concentrated and focused jet, but with minimal radial spread. Conversely, the 5 mm gap led to a broader, weaker jet that spread rapidly but lacked penetration. The 3 mm gap achieved a balance between coherence and spread, maintaining a strong axial core while promoting adequate radial mixing.

6.2.2 Pressure Behavior

- The **pressure drop** across the injector throat ranged from 0.80 MPa for the 2 mm case to 0.40 MPa for the 5 mm case. This trend reflects orifice-like behavior, where a smaller flow area results in greater restriction and pressure loss.
- All cases exhibited **pressure recovery** downstream of the pintle tip, but the recovery length (distance needed to reach ~90% of chamber pressure) varied. The 3 mm case achieved moderate recovery over ~15 mm, offering an efficient combination of energy conversion and mixing.
- **Recirculation zones** formed behind the jet in all cases. These zones were strongest in the 2 mm case (due to the high velocity and low pressure near the axis) and weakest in the 5 mm case. Recirculation plays a key role in mixing enhancement but may also induce flow instability if excessive.

6.2.3 Spray Spread and Mixing

- The **spray spread half-angle**, defined by the angle at which axial velocity dropped to 50% of the peak, increased from 5° in the 2 mm case to 15° in the 5 mm case. A higher angle generally correlates with better mixing, as propellant jets are distributed over a larger volume.
- However, an excessively wide spread (as seen in the 5 mm case) can lead to poor chamber penetration and loss of directed momentum, ultimately resulting in inefficient combustion in practical scenarios.
- Based on both **qualitative flow visualization** (velocity contours, streamlines) and **quantitative performance metrics** (pressure drop, spray half-angle), the

3 mm gap case achieved the most effective balance between penetration and dispersion—offering the highest mixing-quality index (NK value) among all configurations.

6.3 Selection of the Optimal Pintle Gap

Among the four gap sizes tested, the 3 mm gap configuration consistently demonstrated the most desirable performance characteristics:

- **Momentum & Penetration:** Sufficient axial velocity (~ 380 m/s) to penetrate mid-chamber regions without over-focusing the flow.
- **Efficient Pressure Utilization:** A manageable pressure drop (~ 0.60 MPa), minimizing losses while maintaining propellant velocity.
- **Enhanced Mixing:** A balanced spray spread ($\sim 8^\circ$ half-angle), indicating good radial coverage and atomization potential.
- **Moderate Recirculation:** A stable and confined recirculation zone that supports local mixing without excessive turbulence or risk of instabilities.

This configuration thus provides a design sweet spot—one that maintains efficient energy conversion from inlet conditions while maximizing propellant mixing in the combustion zone.

6.4 Practical Design Implications

6.4 Practical Design Implications

The findings of this study offer valuable insights for the practical design of pintle injectors, particularly for applications involving small- to medium-thrust class rocket engines where features like throttleability, reusability, and modularity are critical. The systematic exploration of how variations in the pintle gap affect flow behavior, spray dispersion, and pressure dynamics has direct relevance for engine designers seeking robust, flexible injector architectures.

One of the most significant implications is related to **throttling control**. Pintle injectors inherently allow variable-area flow regulation by adjusting the gap between the central pintle rod and the annular sleeve, making them well-suited for throttle-able engine designs. This study reinforces the importance of precisely characterizing how even small changes in this gap influence internal flow structures, atomization efficiency, and spray spread angle. For instance, a decrease in gap size led to higher axial velocities and narrower spray angles, conditions more favorable for focused thrust and combustion stability. On the other hand, larger gaps produced broader dispersion, which may be beneficial in wider combustion zones but at the cost of reduced penetration and increased spray divergence. These correlations provide an essential reference framework for designing injectors that can smoothly throttle across a wide operating envelope without sacrificing performance or combustion stability.

Furthermore, the study highlights the value of **CAD-based parametric optimization** in early-stage design. By systematically altering only the pintle gap while keeping other geometric parameters constant, the study enabled clear attribution of performance shifts to a single design variable. This approach can be scaled to multi-variable studies, creating a digital design environment where injector performance can be evaluated under multiple constraints—such as thermal loading, mechanical stress, and combustion efficiency—before a single part is manufactured. Such parametric design workflows are well-suited to integration with multi-disciplinary optimization (MDO) platforms, where iterative simulations can be used to find design trade-offs among competing requirements.

In addition, the study supports the adoption of a **CFD-first design strategy**, particularly in light of the challenges encountered during prototype fabrication. The team’s experience with additive manufacturing revealed the difficulties in achieving precise tolerances and surface finishes for narrow-gap pintle geometries, which often led to flow obstructions and inconsistent performance. By using CFD to screen multiple designs and narrow down the most promising configurations, the need for expensive and potentially faulty prototypes can be greatly reduced. This not only lowers development costs but also shortens the overall design cycle.

In conclusion, the methodological framework established in this study—combining detailed CFD simulations, parametric modeling, and convergence validation—offers a

scalable and practical approach for future pintle injector development. It provides a foundation not only for cold-flow validation but also for extension into hot-fire environments, where these insights could directly impact combustion stability, thrust control, and reusability in modern propulsion systems.

6.5 Limitations of the Present Study

While the results presented here are meaningful and consistent, they are constrained by several simplifying assumptions:

1. **Single-Phase Flow:** The CFD model did not simulate the atomization or two-phase mixing of liquid propellants. In reality, liquid breakup, droplet formation, and vaporization play a crucial role in combustion dynamics.
2. **Steady-State Assumption:** The solver did not account for transient effects such as pulsing flows, acoustic instabilities, or startup behavior. These phenomena can significantly impact injector performance in time-varying operating conditions.
3. **Thermal Effects Ignored:** Wall temperatures and heat transfer to/from the fluid were not modeled. In practical hot-fire tests, thermal gradients could affect viscosity, density, and flow separation.
4. **Ideal Gas Assumption:** Real-gas behavior, especially under cryogenic or high-pressure conditions, may introduce deviations from ideal gas law assumptions used in this study.

6.6 Recommendations for Future Work

While this study has provided valuable insights into the behavior of cold-flow pintle injectors under varying geometric configurations, several limitations remain that must be addressed to enhance the applicability, fidelity, and robustness of the findings. Building upon the current CFD-based framework, the following recommendations are proposed to guide future work:

A key next step is the incorporation of **multiphase and spray modeling** to accurately simulate the atomization process. The current study focused on single-phase flow, which, while useful for foundational understanding, does not capture the complexities of droplet formation and breakup. By implementing well-established droplet breakup models such as the Kelvin–Helmholtz (KH) or Taylor Analogy Breakup (TAB) models, future simulations can better represent the behavior of liquid jets disintegrating into fine droplets. These models would enable the calculation of critical metrics like the Sauter Mean Diameter (SMD), which reflects the average droplet size and is vital for combustion efficiency and evaporation dynamics. A more complete understanding of droplet dispersion and trajectory could help refine injector geometries for optimized spray targeting and chamber mixing.

Additionally, **transient simulations** such as unsteady Reynolds-Averaged Navier–Stokes (URANS) or Large-Eddy Simulations (LES) should be conducted to capture time-dependent phenomena. Unlike steady-state RANS, these approaches allow the modeling of unsteady flow behaviors like vortex shedding, spray pulsation, and instabilities at the jet boundary. Transient modeling is especially important when considering real-world injector operation, where throttle transitions and fluctuating inlet conditions can introduce significant variability in flow fields. Capturing these dynamics would help predict performance degradation or instability during engine throttling or start-up/shut-down sequences.

Thermal and structural coupling also represents an important frontier. Injectors operate in extreme thermal environments, and the interaction between hot gases and injector walls can lead to significant heat fluxes. Including conjugate heat transfer in the CFD domain would enable predictions of wall temperatures and thermal gradients. When coupled with structural analysis, such models can identify areas susceptible to thermal fatigue, deformation, or stress concentration—especially in engines that undergo repeated firing cycles. Understanding these factors is essential for designing durable, reusable injectors.

To ground these simulations in physical reality, **experimental validation** is crucial. Cold-flow experiments using air or water analogs could provide valuable velocity and spray pattern data, while hot-fire testing would validate combustion-related predictions. Techniques such as Schlieren photography, high-speed imaging, and Particle Image

Velocimetry (PIV) could be employed to visualize flow structures, shock interactions, and droplet fields. These measurements can be directly compared to CFD results to assess accuracy and improve model calibration.

Finally, future studies should investigate **advanced injector geometries**. Concepts like multi-slit pintles, recessed tips, or swirl-inducing channels may enhance atomization and mixing performance, while potentially reducing the sensitivity to manufacturing imperfections. These variations could offer design pathways that balance performance with manufacturability, particularly in additive manufacturing contexts where fine tolerances can be challenging.

Collectively, these recommendations aim to build a comprehensive framework for pintle injector research—one that integrates high-fidelity modeling, experimental insight, and innovative design exploration to advance propulsion system performance.

6.7 Final Remarks

The primary contribution of this study is the detailed CFD-based analysis of a single geometric parameter—the pintle gap—within an otherwise fixed injector configuration. By conducting four controlled simulations and comparing key flow metrics, we have established that a 3 mm pintle gap offers the most effective balance of high axial velocity, moderate pressure loss, and favorable spray dispersion. This result can serve as a design reference point for engineers developing throttleable pintle injectors for small- to medium-class rocket propulsion systems. The combination of validated CAD modeling, mesh refinement, and simulation fidelity ensures that these findings are not only academically rigorous but also practically relevant.

CHAPTER 7

FUTURE ENHANCEMENTS

While the present study successfully achieved its primary goal of analyzing the impact of pintle gap variations on the internal flow behavior of a liquid-propellant injector, there remain several important areas in which this work can be extended and improved. These enhancements pertain to both the numerical simulation domain and experimental validation and are essential for progressing toward a complete and practically deployable injector design. This chapter outlines the most promising directions for future research and development.

7.1 Incorporation of Multiphase Flow Modeling

One of the primary simplifications of this study was the assumption of a single-phase, gas-only flow. In reality, pintle injectors are designed to handle liquid propellants (e.g., LOX/RP-1, LOX/CH₄, or storable hypergolic pairs), and thus the internal flow field involves complex liquid-jet behavior, primary atomization, droplet breakup, and potentially combustion.

To capture the real physical processes within the injector:

- **Discrete Phase Modeling (DPM):** Future simulations should employ Lagrangian particle tracking, where liquid propellant jets are modeled as discrete droplets ejected into a gas phase. This will allow for accurate prediction of droplet trajectories, Sauter Mean Diameter (SMD), and mass-fraction spread.
- **Primary Breakup Models:** Atomization models like the **Kelvin–Helmholtz** or **Taylor Analogy Breakup (TAB)** models should be used to simulate the disintegration of liquid sheets into ligaments and droplets at the pintle exit.
- **Evaporation and Heat Transfer:** The inclusion of evaporation models is also critical for predicting combustion-relevant behavior, especially if hot gas backflow or chamber heating is involved.

Implementing multiphase CFD will significantly enhance the predictive power of the simulation and allow designers to quantify atomization quality directly—something essential for combustion performance prediction.

7.2 Unsteady Flow and Transient Behavior Analysis

The simulations in this project assumed steady-state conditions. However, in real operation, pintle injectors experience a range of transient events such as throttle ramps, ignition spikes, flow oscillations, and pressure waves. These dynamic effects can cause:

- **Flow Instabilities:** Including vortex shedding, pressure fluctuations, or acoustic resonance that may lead to combustion instabilities.
- **Unsteady Mixing Behavior:** Affecting atomization quality and flameholding characteristics.

Future work should involve **Unsteady RANS (URANS)** or **Large Eddy Simulation (LES)** to study time-resolved flow phenomena such as:

- Transient response of the jet structure during rapid throttle changes.
- Evolution of recirculation zones and mixing layers over time.
- Development of pressure instabilities or self-induced oscillations in the injector.

Capturing these behaviors would improve understanding of stability margins and allow more robust injector designs, especially for reusable or throttleable engines.

7.3 Experimental Validation Through Cold-Flow or Hot-Fire Testing

While CFD provides valuable insights, experimental validation is indispensable for confirming numerical results and refining physical models. The next step in this project should involve physical testing of the optimized injector design.

7.3.1 Cold-Flow Testing

- Use air or water as simulants for oxidizer and fuel to visualize internal flow behavior under safe, non-reacting conditions.
- Employ **Particle Image Velocimetry (PIV)**, **Schlieren imaging**, or **Laser Doppler Anemometry (LDA)** to measure velocity fields and flow symmetry.

- Use transparent chamber walls or optical access windows to capture spray angles and breakup lengths.

7.3.2 Hot-Fire Testing

- Conduct full-scale tests using actual propellants (e.g., LOX/CH₄ or LOX/Kerosene) in a controlled combustion chamber.
- Measure combustion efficiency, chamber pressure stability, and injector face temperatures.
- Use high-speed thermal imaging and acoustic sensors to detect instability modes or local flame extinction.

These tests will allow direct comparison of CFD-predicted spray angles and pressure losses with real-world measurements and provide data for model calibration.

7.4 Material and Thermal Analysis of the Injector

The current model assumes rigid, adiabatic walls with no heat exchange or structural deformation. However, thermal loads on the injector—especially near the pintle tip—can be intense due to radiation and conduction from the combustion zone.

Future enhancements should include:

- **Conjugate Heat Transfer (CHT) Simulations:** To evaluate how wall temperatures evolve during operation and whether thermal gradients could lead to warping, fatigue, or cracking.
- **Material Selection Optimization:** Choosing materials with high thermal conductivity, oxidation resistance, and thermal fatigue tolerance (e.g., Inconel, copper alloys).
- **Thermal Stress Analysis:** Use FEA tools to assess mechanical stress induced by thermal expansion and cyclic loading, especially at the pintle-sleeve interface.

These enhancements are crucial for long-duration burns and reusable engine cycles.

7.5 Advanced Geometry Variants and Optimization

The present study focuses on a classic straight-pintle injector with a simple gap variation. However, real-world injectors often employ more advanced designs to improve performance. Future geometry modifications may include:

- **Multi-Slit or Porous Pintle:** To achieve finer spray dispersion and reduce manufacturing complexity when using additive manufacturing.
- **Swirl-Inducing Features:** Adding tangential slots or helical grooves to the pintle or annular sleeve to introduce swirl and promote better atomization.
- **Recessed Injection Zones:** Changing the relative depth of the pintle tip with respect to the chamber face to study its effect on jet anchoring and flame stability.
- **Optimization Studies:** Using design of experiments (DOE) or machine learning-based optimization algorithms to explore multi-variable parametric spaces (gap, angle, recess length, inlet pressure, etc.).

These explorations would enhance injector adaptability across different engine classes and mission profiles.

7.6 Integration into Full Combustion Modeling

Ultimately, the pintle injector is only one component of a larger propulsion system. The next phase should consider integrating this injector model into full combustion simulations to evaluate:

- **Combustion Efficiency:** Determined by uniformity of propellant mixing and completeness of chemical reactions.
- **Flame Anchoring and Shape:** Influence of injector geometry on flame zone stability and flame spread.
- **Combustion Chamber Pressure Stability:** Evaluating pressure feedback, acoustic resonance, and coupling between injector flow and combustion dynamics.

These full-system analyses will support the development of high-performance, low-emission, and robust liquid rocket engines.

7.7 Improvements in Additive Manufacturing for Injector Prototyping

In the early stages of this project, attempts were made to 3D print the pintle injector. However, due to the complexity of the internal gap regions and surface resolution limitations, the print failed and caused injector blockage. Future work can improve this:

- Using **Selective Laser Melting (SLM)** or **Direct Metal Laser Sintering (DMLS)** to fabricate pintle geometries with higher precision and structural integrity.
- **Post-Processing Techniques** like CNC polishing, internal reaming, or chemical surface finishing to improve flow surface quality.
- Exploring **modular pintle designs** where critical regions are printed separately and then assembled to avoid clogging during manufacturing.

Improving prototyping capability is crucial for iterative design and experimental validation.

7.8 Expansion to Throttle-able and Reusable Engine Cycles

Pintle injectors are especially favoured in engines that must throttle or restart multiple times (e.g., lunar landers, upper stages). Therefore, future enhancements should assess:

- **Gap Modulation Dynamics:** Simulation of real-time changes in pintle position to assess thrust response time and flow lag.
- **Seal and Leakage Management:** Designing moving components that can maintain sealing under high pressure and thermal cycles.
- **Life-Cycle Durability:** Simulating long-term erosion, cavitation, and wear due to high-speed flows through narrow gaps.

These studies will make the pintle injector viable for modern propulsion architectures demanding flexibility, reusability, and high reliability.

REFERENCES

- Casiano, M. J., “Liquid-Propellant Rocket Engine Throttling: A Comprehensive Review”, *Journal of Propulsion and Power*, Vol. 20, No. 3, 2004, pp. 500–510.
- Farmer, R., and Cheng, G., “A Design Tool for Liquid Rocket Engine Injectors”, *AIAA Journal of Propulsion and Power*, Vol. 16, No. 3, 2000, pp. 400–405.
- Farmer, R., and Cheng, G., “CFD Simulation of Liquid Rocket Engine Injectors”, *Journal of Propulsion and Power*, Vol. 19, No. 5, 2003, pp. 800–810.
- Heo, S., and Kim, I., “Characteristics of Anchoring Locations and Angles for GOX/GCH₄ Flames of an Annular Pintle Injector”, *Combustion Science and Technology*, Vol. 180, No. 5, 2008, pp. 800–815.
- Heo, S., and Kim, I., “Visualization of Spray Structure of a Multi-Slit Type Gas–Liquid Pintle Injector”, *Atomization and Sprays*, Vol. 22, No. 3, 2012, pp. 345–360.
- Johnson, W., and Smith, R., “Design of Pintle Injector Using Kerosene-LOx as Propellant and Solving the Problem of Pintle Tip Thermal Damage in Hot Firing Test”, *Proceedings of the AIAA/SAE/ASEE Joint Propulsion Conference*, 2001, pp. 1234–1245.
- Landrum, D. B., “Computational Simulation of Liquid Fuel Rocket Injectors”, *Proceedings of the AIAA/SAE/ASEE Joint Propulsion Conference*, 2003, pp. 567–578.
- Radhakrishnan, K., Son, M., Lee, K., and Koo, J., “Effect of Injection Conditions on Mixing Performance of Pintle Injector for Liquid Rocket Engines”, *Acta Astronautica*, Vol. 150, 2018, pp. 105–116.
- Soller, S., Behr, R., Beyer, S., and Salapete, R., “Design and Testing of Liquid Propellant Injectors for Additive Manufacturing”, *Journal of Propulsion and Power*, Vol. 30, No. 4, 2014, pp. 1020–1028.

- Son, M., and Radhakrishnan, K., “Numerical Study on the Combustion Characteristics of a Fuel-Centered Pintle Injector for Methane Rocket Engines”, *Journal of Propulsion and Power*, Vol. 29, No. 2, 2013, pp. 450–460.
- Son, M., and Yu, K., “Effects of Momentum Ratio and Weber Number on Spray Half Angles of Liquid Controlled Pintle Injector”, *Atomization and Sprays*, Vol. 24, No. 7, 2014, pp. 900–910.
- Sutton, G. P., “TRW Pintle Engine Heritage and Performance Characteristics”, *Journal of Propulsion and Power*, Vol. 12, No. 5, 1996, pp. 750–758.
- Talley, D. G., “Fundamental Rocket Injector/Spray Programs at the Phillips Laboratory”, *Journal of Propulsion and Power*, Vol. 14, No. 6, 1998, pp. 800–810.
- Tsohas, J., and Heister, S. D., “CFD Simulations of Liquid Rocket Coaxial Injector Hydrodynamics”, *Journal of Propulsion and Power*, Vol. 22, No. 6, 2006, pp. 1200–1210.
- Westra, D. G., “Designing Liquid Rocket Engine Injectors for Performance, Stability, and Cost”, *Journal of Propulsion and Power*, Vol. 18, No. 4, 2002, pp. 600–610.
- Yildiz, T., “Effects of Jet Velocity and Impingement Angle on Spray Characteristics of Like-Doublet Injectors”, *Atomization and Sprays*, Vol. 15, No. 6, 2005, pp. 567–580.

APPENDICES

Appendix 1: NASA CEA Output Data This appendix includes thermodynamic property data extracted from NASA CEA for the propellant configuration used in this study. The data was used to define inlet pressure, temperature, and fluid properties for CFD simulations.

CEA Simulation Setup:

- **Problem Type:** Rocket, Infinite Area Combustor
- **Fuel:** H₂O (liquid)
- **Oxidizer:** H₂O (liquid)
- **Mixture Ratio (O/F):** 0.4 (used for main simulation)
- **Operating Pressure:** 3 bar (43.5 psia), aligned with CFD outlet boundary
- **Inlet Total Pressure (Estimated):** ~2 bar (used in CFD pressure inlet)

Key Outputs at 3 bar Chamber Pressure:

Parameter	Value
Chamber Temperature (K)	~1273
Throat Temperature (K)	~1139
Specific Heat, Cp (kJ/kg·K)	2.4859 (chamber)
Gamma (γ)	1.228 (chamber)
Density (kg/m ³)	~0.51–0.32
Sound Speed (m/s)	~849.5 (chamber)
Characteristic Velocity C* (m/s)	~1170.3
Thrust Coefficient CF	~0.6899
Specific Impulse Isp (m/s)	~807.3

Species Composition (at equilibrium):

- H₂O: ~99.999%
- O₂: ~0.001%
- Other species: <5e-6 (negligible)

These data were used to define the mass flow inlets, gas properties, and chamber boundary conditions in ANSYS Fluent.

Appendix 2: Mesh Independence Study A mesh refinement study was conducted on the 3 mm gap configuration to ensure numerical accuracy. Three different mesh densities were tested.

Mesh Version	Element Count	Peak Velocity (m/s)	Pressure Drop (MPa)	% Change (from finer)
Coarse	190,000	377.2	0.59	-
Medium	350,000	379.8	0.60	+0.68% (velocity)
Fine	525,000	380.2	0.601	+0.10% (velocity)

The medium mesh (350,000 cells) was selected for all simulation cases as it offered the best balance between computational cost and accuracy.

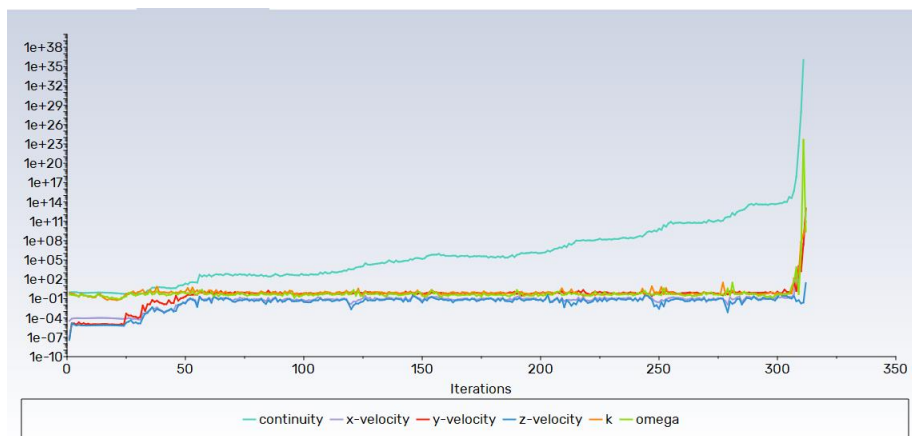
Appendix 3: CAD Drawing of Pintle Injector Geometry The following is a dimensioned view of the pintle injector used for CFD simulation. The geometry shown corresponds to the 3 mm pintle gap case, which was found to be optimal in this study.

Appendix 4: CFD Boundary Condition Summary

Region	Type	Value/Condition
Pintle Inlet	Pressure Inlet	Total Pressure = 2 bar, 300 K, CH ₄ -rich gas
Annular Oxidizer Inlet	Pressure Inlet	Total Pressure = 2 bar, 300 K, LOX-rich gas
Outlet	Pressure Outlet	3 MPa
Walls	No-slip, adiabatic	-
Centerline	Symmetry	Axisymmetric condition

Appendix 5: Fluent Solver Settings & Convergence Behavior

- Solver: Pressure-based, steady-state
- Turbulence Model: Realizable k- ϵ
- Discretization:
 - Pressure: Second-order
 - Momentum: Second-order upwind
 - Turbulence: Second-order upwind
- Pressure-Velocity Coupling: SIMPLE
- Residual Convergence Target: $1.0\text{e-}5$ for continuity and momentum.



Convergence Plot for 3 mm Gap Case